Q1] CREATE DATABASE Restaurant;
USE Restaurant;

```
CREATE TABLE Category(
category_id INT,
category_name VARCHAR(50)
);

CREATE TABLE Dish(
dish_id INT,
dish_name
VARCHAR(50),
category_id INT, price
DECIMAL(10,2),
dish_rank VARCHAR(50)
);

    CREATE TABLE Customer(
customer_id INT,
customer_name VARCHAR(50),
phone VARCHAR(50)
);

    CREATE TABLE Orders(
order_id INT,
customer_id INT,
order_date DATE
);

    CREATE TABLE OrderSpecs(
specs_id INT,
order_id INT,
dish_id INT,
quantity INT
);
```

Q2] ALTER TABLE Category ADD PRIMARY KEY(category_id);
ALTER TABLE Dish ADD PRIMARY KEY(dish_id);
ALTER TABLE Customer ADD PRIMARY KEY(customer_id);
ALTER TABLE Orders ADD PRIMARY KEY(order_id);
ALTER TABLE OrderSpecs ADD PRIMARY KEY(specs_id);

Q3] INSERT INTO Category Values
(1,"Chinese"),(2,"Indian"),(3,"Punjabi"),(4,"Italian"),(5,"Afgani");

INSERT INTO Dish Values
(101,"Veg Noodles",1,160.00,"Regular"),
(102,"Veg Special Thali",1,200.00,"Popular"),
(103,"Dal Makhani",2,180.00,"Special"),
(104,"Pasta",3,360.00,"Popular"),
(105,"Chicken Afgani",4,260.00,"Regular");

INSERT INTO Customer Values
(1,"Sai","9090796659"),
(2,"Raj","9093079659"),
(3,"Neha","9019079659"),
(4,"Amit","9209079659"),
(5,"Rahul","9009079659"),

INSERT INTO Orders Values
(201,1,"2026-10-11"),
(202,2,"2026-10-12"),
(203,3,"2026-10-13"),
(204,4,"2026-10-14"),
(205,5,"2026-10-15");

INSERT INTO OrderSpecs Values
(1,201,101,2),
(2,202,102,1),
(3,203,103,3),
(4,204,104,1),
(5,205,105,2);

Q4] ALTER TABLE Dish
ADD FOREIGN KEY(category_id) REFERENCES Category(category_id);

ALTER TABLE Orders
ADD FOREIGN KEY(customer_id) REFERENCES Customer(customer_id);

ALTER TABLE OrderSpecs
ADD FOREIGN KEY(order_id) REFERENCES Orders(order_id)

```
Q5]   SELECT * FROM Dish
      WHERE price < 250 OR category_id=4;


Q6]   SELECT dish_name,category_id,price,
 RANK() OVER(ORDER BY price DESC) as dish_rank
 FROM Dish;



Q7] SELECT category_id, COUNT(*) AS total_dishes
       FROM Dish
       GROUP BY category_id

Q8] SELECT dish_name,price
    FROM Dish
    WHERE price=(SELECT MAX(price) FROM Dish);


Q9] CREATE VIEW DishView AS
    SELECT dish_id,dish_name,price
    FROM Dish;

    SELECT * FROM DishView;

    UPDATE DishView
    SET price=240.00
    WHERE dish_id=101;

    SELECT * FROM Dish
    WHERE dish_id=101;



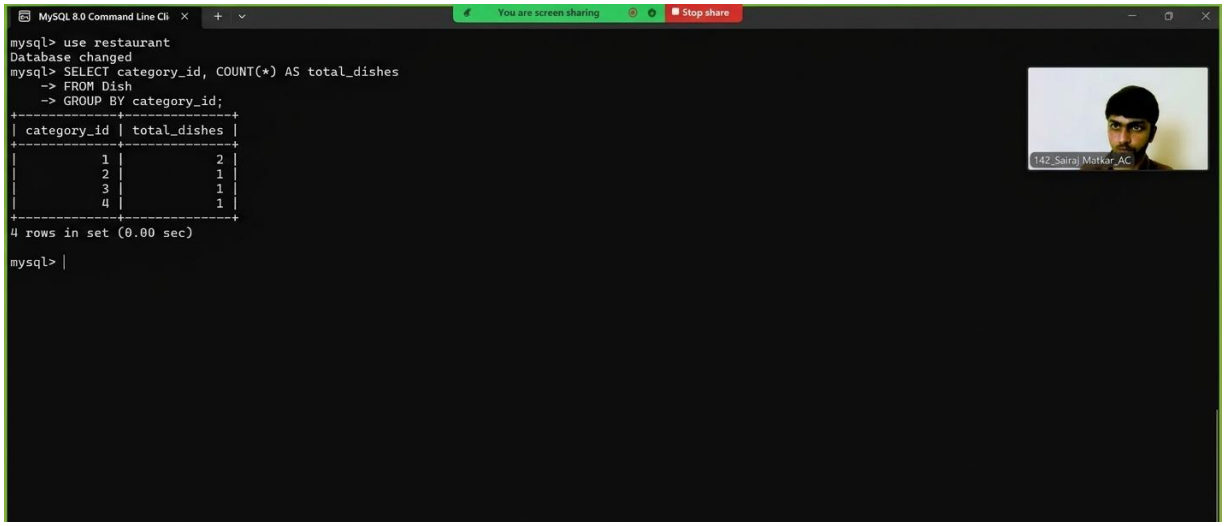
Q10]  SELECT d.dish_name,c.category_name
      FROM Dish d       LEFT
JOIN Category c ON
      d.category_id=c.category_id;

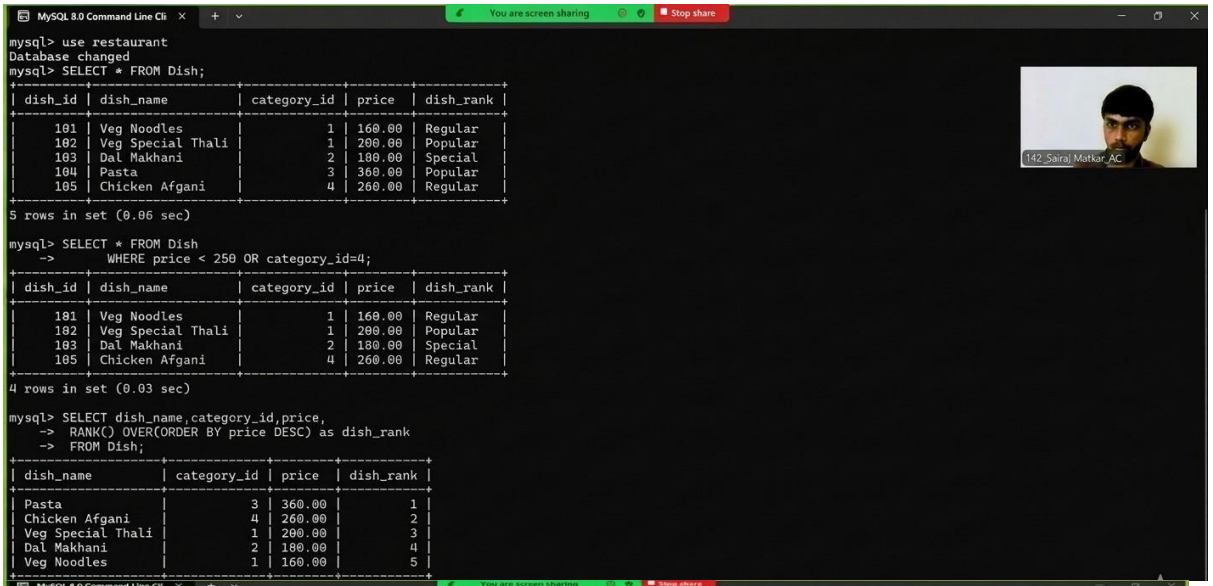

Q11]  CREATE TEMPORARY TABLE ChineseMenu AS
      SELECT dish_id, dish_name, price
      FROM Dish
      WHERE category_id = 1;
```

SELECT * FROM ChineseMenu;

DROP TEMPORARY TABLE ChineseMenu;

Q12] CREATE PROCEDURE CountDishesByCat(
IN p_category

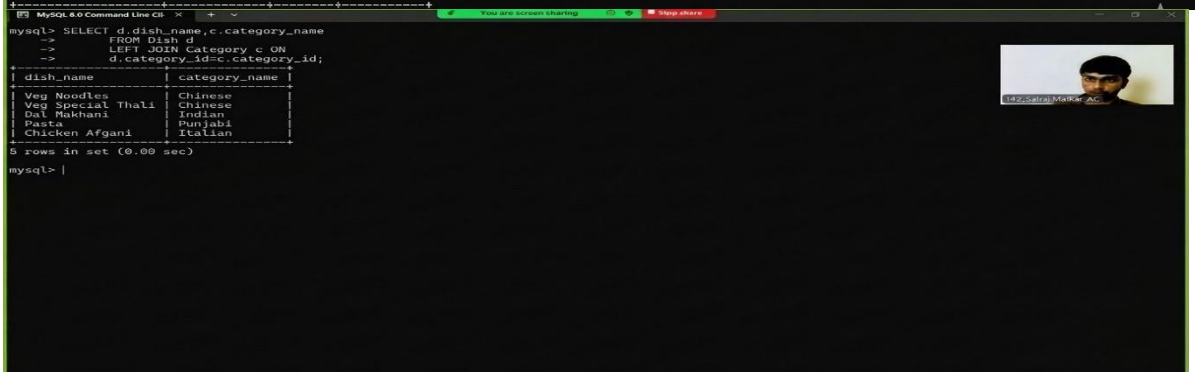




```
mysql> use restaurant
Database changed
mysql> SELECT * FROM Dish;
+---------+-------------------+-------------+--------+-----------+
| dish_id | dish_name         | category_id | price  | dish_rank |
+---------+-------------------+-------------+--------+-----------+
|     101 | Veg Noodles       |           1 | 160.00 | Regular   |
|     102 | Veg Special Thali |           1 | 200.00 | Popular   |
|     103 | Dal Makhani       |           2 | 180.00 | Special   |
|     104 | Pasta             |           3 | 360.00 | Popular   |
|     105 | Chicken Afgani    |           4 | 260.00 | Regular   |
+---------+-------------------+-------------+--------+-----------+
5 rows in set (0.06 sec)

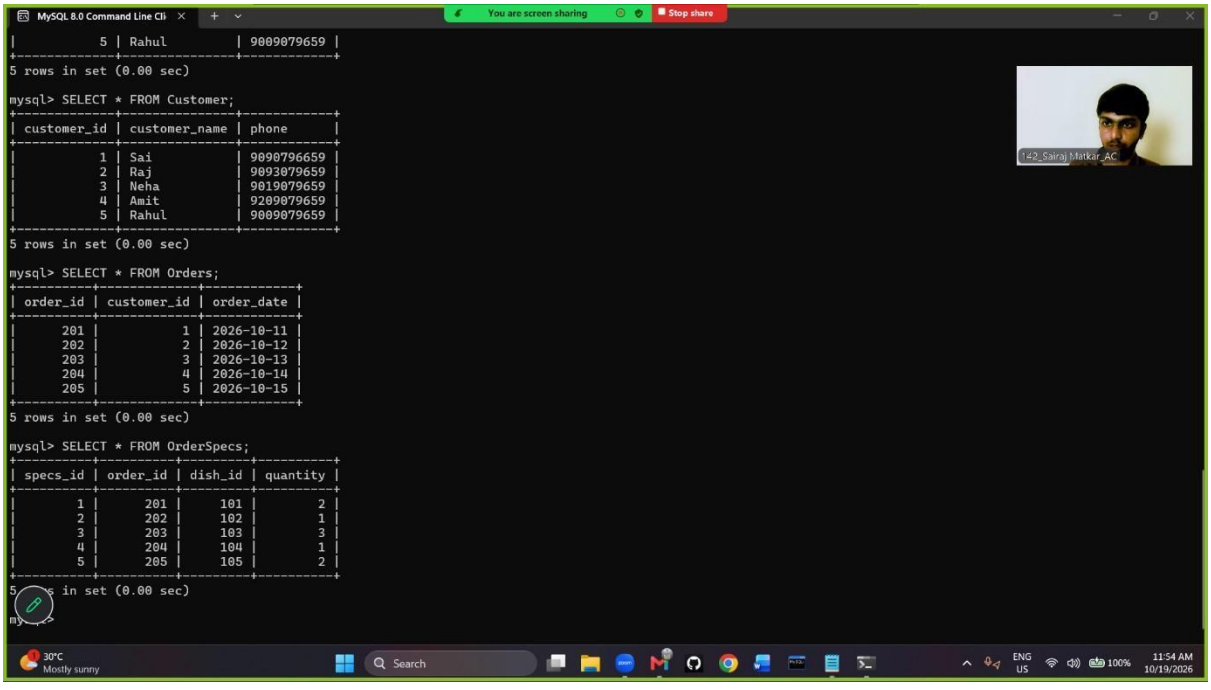
mysql> SELECT * FROM Dish
    ->         WHERE price < 250 OR category_id=4;
+---------+-------------------+-------------+--------+-----------+
| dish_id | dish_name         | category_id | price  | dish_rank |
+---------+-------------------+-------------+--------+-----------+
|     101 | Veg Noodles       |           1 | 160.00 | Regular   |
|     102 | Veg Special Thali |           1 | 200.00 | Popular   |
|     103 | Dal Makhani       |           2 | 180.00 | Special   |
|     105 | Chicken Afgani    |           4 | 260.00 | Regular   |
+---------+-------------------+-------------+--------+-----------+
4 rows in set (0.03 sec)

mysql> SELECT dish_name,category_id,price,
    ->  RANK() OVER(ORDER BY price DESC) as dish_rank
    ->  FROM Dish;
+-------------------+-------------+--------+-----------+
| dish_name         | category_id | price  | dish_rank |
+-------------------+-------------+--------+-----------+
| Pasta             |           3 | 360.00 |         1 |
| Chicken Afgani    |           4 | 260.00 |         2 |
| Veg Special Thali |           1 | 200.00 |         3 |
| Dal Makhani       |           2 | 180.00 |         4 |
| Veg Noodles       |           1 | 160.00 |         5 |
+-------------------+-------------+--------+-----------+
```



```
mysql> SELECT d.dish_name,c.category_name
    ->         FROM Dish d
    ->         LEFT JOIN Category c ON
    ->         d.category_id=c.category_id;
+-------------------+---------------+
| dish_name         | category_name |
+-------------------+---------------+
| Veg Noodles       | Chinese       |
| Veg Special Thali | Chinese       |
| Dal Makhani       | Indian        |
| Pasta             | Punjabi       |
| Chicken Afgani    | Italian       |
+-------------------+---------------+
5 rows in set (0.00 sec)

mysql>
```



```
|           5 | Rahul          | 9009079659 |
+-------------+----------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Customer;
+-------------+---------------+------------+
| customer_id | customer_name | phone      |
+-------------+---------------+------------+
|           1 | Sai           | 9090796659 |
|           2 | Raj           | 9093079659 |
|           3 | Neha          | 9019079659 |
|           4 | Amit          | 9209079659 |
|           5 | Rahul         | 9009079659 |
+-------------+---------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Orders;
+----------+-------------+------------+
| order_id | customer_id | order_date |
+----------+-------------+------------+
|      201 |           1 | 2026-10-11 |
|      202 |           2 | 2026-10-12 |
|      203 |           3 | 2026-10-13 |
|      204 |           4 | 2026-10-14 |
|      205 |           5 | 2026-10-15 |
+----------+-------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM OrderSpecs;
+----------+----------+---------+----------+
| specs_id | order_id | dish_id | quantity |
+----------+----------+---------+----------+
|        1 |      201 |     101 |        2 |
|        2 |      202 |     102 |        1 |
|        3 |      203 |     103 |        3 |
|        4 |      204 |     104 |        1 |
|        5 |      205 |     105 |        2 |
+----------+----------+---------+----------+
5 rows in set (0.00 sec)
```

```
    ->          (2,202,102,1),
    ->          (3,203,103,3),
    ->          (4,204,104,1),
    ->          (5,205,105,2);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CATEGORY;
+-------------+---------------+
| category_id | category_name |
+-------------+---------------+
|           1 | Chinese       |
|           2 | Indian        |
|           3 | Punjabi       |
|           4 | Italian       |
|           5 | Afgani        |
+-------------+---------------+
5 rows in set (0.01 sec)

mysql> SELECT * FROM Dish;
+---------+------------------+-------------+--------+-----------+
| dish_id | dish_name        | category_id | price  | dish_rank |
+---------+------------------+-------------+--------+-----------+
|     101 | Veg Noodles      |           1 | 160.00 | Regular   |
|     102 | Veg Special Thali |           1 | 200.00 | Popular   |
|     103 | Dal Makhani      |           2 | 180.00 | Special   |
|     104 | Pasta            |           3 | 360.00 | Popular   |
|     105 | Chicken Afgani   |           4 | 260.00 | Regular   |
+---------+------------------+-------------+--------+-----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Customer;
+-------------+---------------+------------+
| customer_id | customer_name | phone      |
+-------------+---------------+------------+
|           1 | Sai           | 9090796659 |
|           2 | Raj           | 9093079659 |
|           3 | Neha          | 9019079659 |
|           4 | Amit          | 9209079659 |
|           5 | Rahul         | 9009079659 |
+-------------+---------------+------------+
```

```
mysql> SELECT dish_name,price
    ->      FROM Dish
    ->      WHERE price=(SELECT MAX(price) FROM Dish);
+------------+--------+
| dish_name  | price  |
+------------+--------+
| Pasta      | 360.00 |
+------------+--------+
1 row in set (0.00 sec)

mysql>
```

```
    ->  customer_name VARCHAR(50),
    ->  phone VARCHAR(50),
    ->  );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ')
' at line 5
mysql>   CREATE TABLE Customer(
    ->  customer_id INT,
    ->  customer_name VARCHAR(50),
    ->  phone VARCHAR(50)
    ->  );
Query OK, 0 rows affected (0.02 sec)

mysql>   CREATE TABLE Orders(
    ->  order_id INT,
    ->  customer_id INT,
    ->  order_date DATE
    ->  );
Query OK, 0 rows affected (0.02 sec)

mysql>         CREATE TABLE OrderSpecs(
    ->  specs_id INT,
    ->          order_id INT,
    ->          dish_id INT,
    ->          quantity INT
    ->  );
Query OK, 0 rows affected (0.02 sec)

mysql> SHOW TABLES;
+----------------------+
| Tables_in_restaurant |
+----------------------+
| category             |
| customer             |
| dish                 |
| orders               |
| orderspecs           |
+----------------------+
5 rows in set (0.02 sec)
```

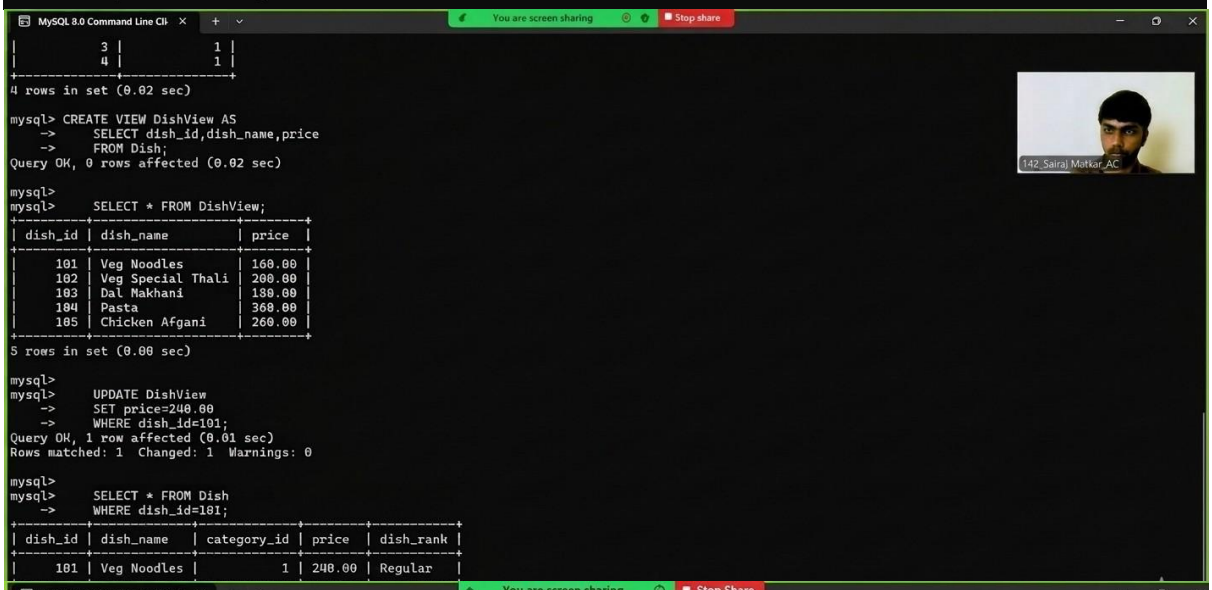


```
MySQL 8.0 Command Line Cli

mysql>  ALTER TABLE Dish
    ->  ADD FOREIGN KEY(category_id) REFERENCES Category(category_id);
Query OK, 5 rows affected (0.06 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>  ALTER TABLE Orders
    ->  ADD FOREIGN KEY(customer_id) REFERENCES Customer(customer_id);
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>  ALTER TABLE OrderSpecs
    ->  ADD FOREIGN KEY(order_id) REFERENCES Orders(order_id);
Query OK, 5 rows affected (0.04 sec)
Records: 5  Duplicates: 0  Warnings: 0
```



```
MySQL 8.0 Command Line Cli
You are screen sharing
Stop share

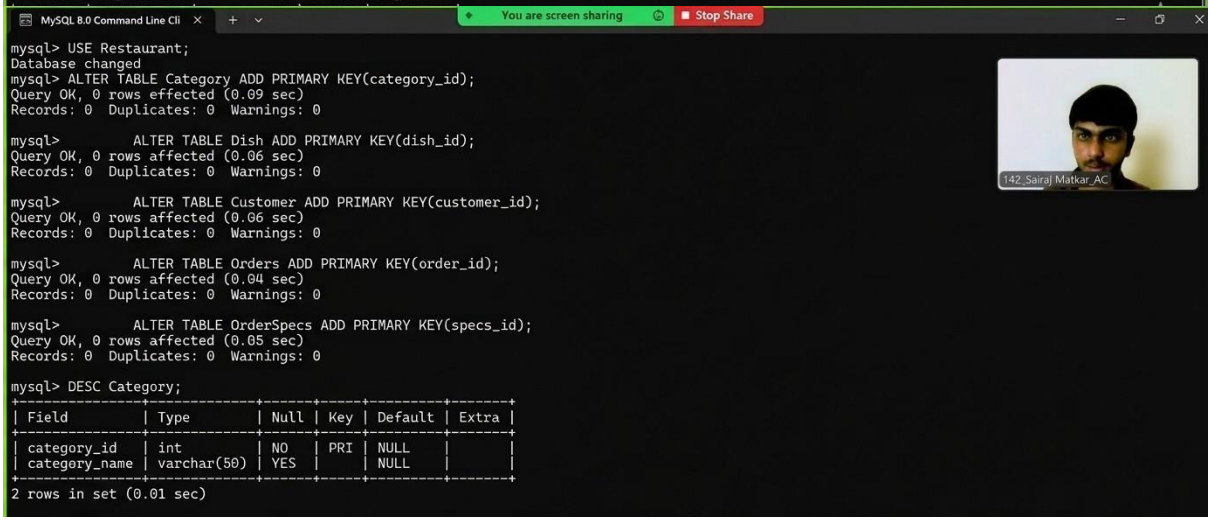
|           3 |           1 |
|           4 |           1 |
+-------------+-------------+
4 rows in set (0.02 sec)

mysql> CREATE VIEW DishView AS
    ->     SELECT dish_id,dish_name,price
    ->     FROM Dish;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql>     SELECT * FROM DishView;
+---------+------------------+--------+
| dish_id | dish_name        | price  |
+---------+------------------+--------+
|     101 | Veg Noodles      | 160.00 |
|     102 | Veg Special Thali | 200.00 |
|     103 | Dal Makhani      | 130.00 |
|     104 | Pasta            | 360.00 |
|     105 | Chicken Afgani   | 260.00 |
+---------+------------------+--------+
5 rows in set (0.00 sec)

mysql>
mysql>     UPDATE DishView
    ->     SET price=240.00
    ->     WHERE dish_id=101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>     SELECT * FROM Dish
    ->     WHERE dish_id=101;
+---------+-----------+-------------+--------+-----------+
| dish_id | dish_name | category_id | price  | dish_rank |
+---------+-----------+-------------+--------+-----------+
|     101 | Veg Noodles |           1 | 240.00 | Regular   |
```

```
MySQL 8.0 Command Line Cli
You are screen sharing
Stop Share

mysql> USE Restaurant;
Database changed
mysql> ALTER TABLE Category ADD PRIMARY KEY(category_id);
Query OK, 0 rows effected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE Dish ADD PRIMARY KEY(dish_id);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE Customer ADD PRIMARY KEY(customer_id);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE Orders ADD PRIMARY KEY(order_id);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE OrderSpecs ADD PRIMARY KEY(specs_id);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Category;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| category_id   | int         | NO   | PRI | NULL    |       |
| category_name | varchar(50) | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```